

Demographic Analysis Using R

Population & Health Data with Tidyverse

Workshop on Demographic Data Analysis

2025-11-11

Welcome to Demographic Data Analysis!

What You'll Learn Today

- Import and explore health/demographic data
- Clean and transform population datasets
- Calculate demographic indicators
- Analyze health outcomes by age, sex, and socioeconomic factors
- Create reproducible demographic reports
- Visualize population health patterns

We'll use real NHANES data - the same type used by demographers and public health researchers worldwide

About NHANES Data

National Health and Nutrition Examination Survey (NHANES)

- Conducted by CDC (USA)
- Representative sample of US population
- Combines interviews + physical examinations
- Gold standard for health surveillance

Today's variables:

- Demographics: Age, Sex, Race, Education
- Anthropometric: Height, Weight, BMI
- Health: Blood Pressure, Diabetes, Physical Activity
- Socioeconomic: Poverty, Insurance

Why R for Demographic Analysis?

R Advantages:

- ✓ Purpose-built for statistics
- ✓ Excellent for survey data
- ✓ Powerful visualization
- ✓ Reproducible reports
- ✓ Free and open source
- ✓ Growing demographic/health community

Common Tasks:

- Population pyramids
- Age-standardization
- Fertility/mortality analysis
- Health inequality assessment
- Survey data analysis
- Cohort studies

Our Analytical Journey Today

flowchart LR

A[Import Data] → B[Explore & Clean]

B → C[Transform Variables]

C → D[Calculate Indicators]

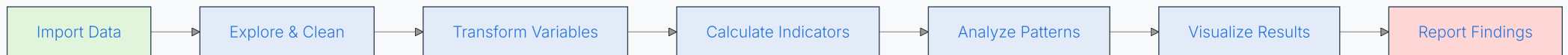
D → E[Analyze Patterns]

E → F[Visualize Results]

F → G[Report Findings]

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Loading Packages

```
library(tidyverse)    # Data wrangling & visualization
library(rio)          # Import/export any format
library(modelsummary) # Summary tables
library(NHANES)       # Our practice dataset
```



Tip

Install packages first if needed:

```
install.packages(c("tidyverse", "rio", "modelsummary", "NHANES"))
```

Loading NHANES Data

```
# Load NHANES dataset
data("NHANES")

# Create a clean working dataset
nhanes_data <- NHANES %>%
  as_tibble() %>%
  # Select relevant variables for demographic analysis
  select(
    # Demographics
    ID, Gender, Age, AgeDecade, Race1, Education,
    MaritalStatus, HomeOwn,
    # Health measures
    Height, Weight, BMI, BPSysAve, BPDiaAve,
    Diabetes, PhysActive, SmokeNow,
    # Socioeconomic
    HHIncome, Poverty
  ) %>%
  # Remove duplicates (NHANES has some repeated IDs)
  distinct(ID, .keep_all = TRUE)
```

First Look at Our Data

```
glimpse(nhanes_data)
```

```
Rows: 6,779
Columns: 18
$ ID      <int> 51624, 51625, 51630, 51638, 51646, 51647, 51654, 51656, ...
$ Gender  <fct> male, male, female, male, male, female, male, male, male...
$ Age     <int> 34, 4, 49, 9, 8, 45, 66, 58, 54, 10, 58, 50, 9, 33, 60, ...
$ AgeDecade <fct> 30-39, 0-9, 40-49, 0-9, 0-9, 40-49, 60-69, 50-59...
$ Race1   <fct> White, Other, White, White, White, White, White, White, ...
$ Education <fct> High School, NA, Some College, NA, NA, College Grad, Som...
$ MaritalStatus <fct> Married, NA, LivePartner, NA, NA, Married, Married, Divo...
$ HomeOwn  <fct> Own, Own, Rent, Rent, Own, Own, Own, Rent, Rent, Own, Re...
$ Height   <dbl> 164.7, 105.4, 168.4, 133.1, 130.6, 166.7, 169.5, 181.9, ...
$ Weight   <dbl> 87.4, 17.0, 86.7, 29.8, 35.2, 75.7, 68.0, 78.4, 74.7, 38...
$ BMI      <dbl> 32.22, 15.30, 30.57, 16.82, 20.64, 27.24, 23.67, 23.69, ...
$ BPSysAve <int> 113, NA, 112, 86, 107, 118, 111, 104, 134, 104, 127, 142...
$ BPDiaAve <int> 85, NA, 75, 47, 37, 64, 63, 74, 85, 68, 83, 68, 63, 74, ...
$ Diabetes <fct> No, No, No, No, No, No, No, No, No, No, No, No, No, No, ...
$ PhysActive <fct> No, NA, No, NA, NA, Yes, Yes, Yes, Yes, NA, Yes, Yes, NA...
$ SmokeNow <fct> No, NA, Yes, NA, NA, NA, No, NA, NA, NA, Yes, NA, NA, No...
$ HHIncome <fct> 25000-34999, 20000-24999, 35000-44999, 75000-99999, 5500...
$ Poverty  <dbl> 1.36, 1.07, 1.91, 1.84, 2.33, 5.00, 2.20, 5.00, 2.20, NA...
```

Data Overview

```
# How many observations?  
nrow(nhanes_data)
```

```
[1] 6779
```

```
# Summary statistics  
datasummary_skim(nhanes_data, histogram = FALSE)
```

	Unique	Missing Pct.	Mean	SD	Min	Median
ID	6779	0	61657.2	5869.4	51624.0	61615.0
Age	81	0	35.5	23.1	0.0	34.0
Height	981	4	160.4	21.1	83.6	165.1

	Unique	Missing Pct.	Mean	SD	Min	Median
Weight	1291	1	69.1	29.9	2.8	71.1
BMI	2074	4	26.5	7.5	12.9	25.8
BPSysAve	127	17	118.0	17.6	76.0	116.0
BPDiaAve	103	17	66.7	14.9	0.0	68.0
Poverty	451	8	2.6	1.7	0.0	2.3
		N	%			
Gender	female	3420	50.4			
	male	3359	49.6			
AgeDecade	0-9	1109	16.4			

Unique		Missing Pct.	Mean	SD	Min	Median
10-19		1016	15.0			
	20-29	880	13.0			
30-39		895	13.2			
	40-49	845	12.5			
50-59		768	11.3			
	60-69	601	8.9			
70+		416	6.1			
Race1	Black	1024	15.1			
Hispanic		495	7.3			

	Unique	Missing Pct.	Mean	SD	Min	Median
	Mexican	859	12.7			
	White	3783	55.8			
	Other	618	9.1			
Education	8th Grade	359	5.3			
	9 - 11th Grade	624	9.2			
	High School	993	14.6			
	Some College	1423	21.0			
	College Grad	1246	18.4			

	Unique	Missing Pct.	Mean	SD	Min	Median
MaritalStatus	Divorced	443	6.5			
	LivePartner	374	5.5			
	Married	2463	36.3			
	NeverMarried	899	13.3			
	Separated	135	2.0			
	Widowed	336	5.0			
HomeOwn	Own	4149	61.2			
	Rent	2419	35.7			
	Other	167	2.5			

	Unique	Missing Pct.	Mean	SD	Min	Median
Diabetes	No	6101	90.0			
	Yes	552	8.1			
PhysActive	No	2473	36.5			
	Yes	2972	43.8			
SmokeNow	No	1092	16.1			
	Yes	964	14.2			
HHIncome	0-4999	159	2.3			
	5000-9999	199	2.9			
	10000-14999	425	6.3			

	Unique	Missing Pct.	Mean	SD	Min	Median
	15000-19999	409	6.0			
	20000-24999	476	7.0			
	25000-34999	689	10.2			
	35000-44999	608	9.0			
	45000-54999	511	7.5			
	55000-64999	390	5.8			
	65000-74999	334	4.9			

Understanding Our Variables

Demographics:

- `Age` : Continuous age in years
- `Gender` : Male/Female
- `Race1` : Race/ethnicity categories
- `Education` : Educational attainment

Health Indicators:

- `BMI`: Body Mass Index (kg/m^2)
- `BPSysAve`: Systolic blood pressure
- `Diabetes`: Yes/No
- `PhysActive`: Physically active (Yes/No)

Socioeconomic:

- `HHIncome`: Household income categories
- `Poverty`: Income-to-poverty ratio

Basic Demographic Profile

```
nhanes_data %>%  
  summarise(  
    n = n(),  
    avg_age = mean(Age, na.rm = TRUE),  
    pct_female = mean(Gender == "female", na.rm = TRUE) * 100,  
    pct_diabetes = mean(Diabetes == "Yes", na.rm = TRUE) * 100  
  )
```

```
# A tibble: 1 × 4  
  n avg_age pct_female pct_diabetes  
<int> <dbl> <dbl> <dbl>  
1  6779  35.5  50.4  8.30
```

Age Distribution

```
nhanes_data %>%  
  count(AgeDecade) %>%  
  mutate(  
    percentage = n / sum(n) * 100,  
    percentage = round(percentages, 1)  
  ) %>%  
  arrange(AgeDecade)
```

```
# A tibble: 9 × 3  
  AgeDecade      n percentage  
  <fct>      <int>      <dbl>  
1 " 0-9"      1109      16.4  
2 " 10-19"    1016       15  
3 " 20-29"     880       13  
4 " 30-39"     895      13.2  
5 " 40-49"     845      12.5  
6 " 50-59"     768      11.3  
7 " 60-69"     601       8.9  
8 " 70+"       416       6.1  
9 <NA>        249       3.7
```

Gender Distribution

```
nhanes_data %>%  
  count(Gender) %>%  
  mutate(  
    percentage = n / sum(n) * 100,  
    percentage = round(percentge, 1)  
  )
```

```
# A tibble: 2 × 3  
  Gender      n percentage  
  <fct> <int>      <dbl>  
1 female  3420      50.4  
2 male    3359      49.6
```


Checking for Missing Data

```
nhanes_data %>%  
  summarise(  
    across(  
      c(Age, Gender, BMI, BPSysAve, Diabetes),  
      ~ sum(is.na(.)) / n() * 100  
    )  
  ) %>%  
  pivot_longer(everything(),  
               names_to = "variable",  
               values_to = "pct_missing") %>%  
  mutate(pct_missing = round(pct_missing, 1))
```

```
# A tibble: 5 × 2  
  variable pct_missing  
  <chr>      <dbl>  
1 Age          0  
2 Gender       0  
3 BMI         4.5  
4 BPSysAve    16.8  
5 Diabetes     1.9
```


Creating Age Groups

```
nhanes_clean <- nhanes_data %>%  
  mutate(  
    # Create standard demographic age groups  
    age_group = case_when(  
      Age < 18 ~ "0-17",  
      Age ≥ 18 & Age < 35 ~ "18-34",  
      Age ≥ 35 & Age < 50 ~ "35-49",  
      Age ≥ 50 & Age < 65 ~ "50-64",  
      Age ≥ 65 ~ "65+",  
      TRUE ~ NA_character_  
    ),  
    # Convert to factor with correct order  
    age_group = factor(age_group,  
                       levels = c("0-17", "18-34", "35-49",  
                                   "50-64", "65+"))  
  )
```

Creating Health Indicators

```
nhanes_clean <- nhanes_clean %>%
  mutate(
    # BMI categories (WHO classification)
    bmi_category = case_when(
      BMI < 18.5 ~ "Underweight",
      BMI ≥ 18.5 & BMI < 25 ~ "Normal",
      BMI ≥ 25 & BMI < 30 ~ "Overweight",
      BMI ≥ 30 ~ "Obese",
      TRUE ~ NA_character_
    ),
    bmi_category = factor(bmi_category,
                        levels = c("Underweight", "Normal",
                                   "Overweight", "Obese")),

    # Hypertension (BP ≥ 140/90)
    hypertension = case_when(
      BPSysAve ≥ 140 | BPDiaAve ≥ 90 ~ "Yes",
      !is.na(BPSysAve) & !is.na(BPDiaAve) ~ "No",
      TRUE ~ NA_character_
    )
  )
```

Creating Socioeconomic Variables

```
nhanes_clean <- nhanes_clean %>%  
  mutate(  
    # Education level (simplified)  
    education_level = case_when(  
      Education %in% c("8th Grade", "9 - 11th Grade") ~ "Less than HS",  
      Education == "High School" ~ "High School",  
      Education == "Some College" ~ "Some College",  
      Education == "College Grad" ~ "College+",  
      TRUE ~ NA_character_  
    ),  
    education_level = factor(education_level,  
                           levels = c("Less than HS", "High School",  
                                       "Some College", "College+")),  
  
    # Poverty status  
    poverty_status = case_when(  
      Poverty < 1 ~ "Below poverty",  
      Poverty ≥ 1 & Poverty < 2 ~ "Near poverty",  
      Poverty ≥ 2 ~ "Above poverty",  
      TRUE ~ NA_character_  
    )  
  )
```

Verifying Our Transformations

```
nhanes_clean %>%  
  select(Age, age_group, BMI, bmi_category,  
         Education, education_level) %>%  
  head(5)
```

```
# A tibble: 5 × 6  
   Age age_group    BMI bmi_category Education education_level  
   <int> <fct>    <dbl> <fct>      <fct>      <fct>  
1    34 18-34    32.2 Obese      High School High School  
2     4 0-17    15.3 Underweight <NA>        <NA>  
3    49 35-49    30.6 Obese      Some College Some College  
4     9 0-17    16.8 Underweight <NA>        <NA>  
5     8 0-17    20.6 Normal    <NA>        <NA>
```


Population Structure by Age and Sex

```
nhanes_clean %>%  
  count(age_group, Gender) %>%  
  group_by(age_group) %>%  
  mutate(  
    total = sum(n),  
    percentage = n / total * 100  
  ) %>%  
  select(age_group, Gender, n, percentage)
```

```
# A tibble: 10 × 4  
# Groups:   age_group [5]  
  age_group Gender      n percentage  
  <fct>     <fct> <int>      <dbl>  
1 0-17      female   939      48.3  
2 0-17      male    1005      51.7  
3 18-34     female   763      50.5  
4 18-34     male    748      49.5  
5 35-49     female   655      50.8  
6 35-49     male    635      49.2  
7 50-64     female   553      49.8  
8 50-64     male    557      50.2  
9 65+       female   510      55.2  
10 65+      male    414      44.8
```


Age-Specific Health Indicators

```
nhanes_clean %>%  
  filter(!is.na(age_group), !is.na(Diabetes)) %>%  
  group_by(age_group) %>%  
  summarise(  
    n = n(),  
    diabetes_prev = mean(Diabetes == "Yes", na.rm = TRUE) * 100,  
    avg_bmi = mean(BMI, na.rm = TRUE),  
    avg_sbp = mean(BPSysAve, na.rm = TRUE)  
  ) %>%  
  mutate(across(where(is.numeric), ~round(., 1)))
```

```
# A tibble: 5 × 5  
  age_group      n diabetes_prev avg_bmi avg_sbp  
  <fct>      <dbl>         <dbl>   <dbl>   <dbl>  
1 0-17      1819            0.4      20     104.  
2 18-34     1511            1.7     27.9    113  
3 35-49     1290            6.2     29.2    118.  
4 50-64     1110           18.9     29.6    125.  
5 65+       923           24.8     28.7    134.
```

Health by Gender

```
nhanes_clean %>%
  group_by(Gender) %>%
  summarise(
    n = n(),
    avg_age = mean(Age, na.rm = TRUE),
    avg_bmi = mean(BMI, na.rm = TRUE),
    diabetes_pct = mean(Diabetes == "Yes", na.rm = TRUE) * 100,
    phys_active_pct = mean(PhysActive == "Yes", na.rm = TRUE) * 100,
    smoker_pct = mean(SmokeNow == "Yes", na.rm = TRUE) * 100
  ) %>%
  mutate(across(where(is.numeric), ~round(., 1)))
```

```
# A tibble: 2 × 7
  Gender      n avg_age avg_bmi diabetes_pct phys_active_pct smoker_pct
  <fct> <dbl>   <dbl>   <dbl>         <dbl>         <dbl>      <dbl>
1 female  3420    36.3    26.9           8           50.7        48
2 male   3359    34.6    26.1          8.6          58.6       46.1
```

Socioeconomic Gradients in Health

```
nhanes_clean %>%  
  filter(!is.na(education_level), !is.na(Diabetes)) %>%  
  group_by(education_level) %>%  
  summarise(  
    n = n(),  
    diabetes_prev = mean(Diabetes == "Yes", na.rm = TRUE) * 100,  
    obesity_prev = mean(bmi_category == "Obese", na.rm = TRUE) * 100,  
    phys_active = mean(PhysActive == "Yes", na.rm = TRUE) * 100  
  ) %>%  
  mutate(across(where(is.numeric), ~round(., 1)))
```

```
# A tibble: 4 × 5  
  education_level      n diabetes_prev obesity_prev phys_active  
  <fct>          <dbl>         <dbl>         <dbl>         <dbl>  
1 Less than HS      982           17.3           38           28.8  
2 High School       993           11.2           37.9          41.2  
3 Some College     1423           11.5           40           54.3  
4 College+         1246            7.9           28.4          71.7
```

BMI Distribution by Age Group

```
nhanes_clean %>%  
  filter(!is.na(age_group), !is.na(bmi_category)) %>%  
  count(age_group, bmi_category) %>%  
  group_by(age_group) %>%  
  mutate(  
    percentage = n / sum(n) * 100,  
    percentage = round(percentage, 1)  
  ) %>%  
  arrange(age_group, bmi_category)
```

```
# A tibble: 20 × 4
```

```
# Groups:   age_group [5]
```

	age_group	bmi_category	n	percentage
	<fct>	<fct>	<int>	<dbl>
1	0-17	Underweight	866	51.3
2	0-17	Normal	551	32.6
3	0-17	Overweight	169	10
4	0-17	Obese	103	6.1
5	18-34	Underweight	50	3.3
6	18-34	Normal	562	37.5
7	18-34	Overweight	426	28.4
8	18-34	Obese	460	30.7
9	35-49	Underweight	13	1
10	35-49	Normal	347	27.1
11	35-49	Overweight	441	34.4
12	35-49	Obese	480	37.5
13	50-64	Underweight	13	1.2
14	50-64	Normal	238	21.6

15	50-64	Overweight	415	37.7
16	50-64	Obese	434	39.5
17	65+	Underweight	13	1.4
18	65+	Normal	246	27.1
19	65+	Overweight	326	35.9

Hypertension Prevalence by Age

```
nhanes_clean %>%
  filter(!is.na(age_group), !is.na(hypertension)) %>%
  group_by(age_group) %>%
  summarise(
    n = n(),
    hypertension_prev = mean(hypertension == "Yes") * 100,
    avg_sbp = mean(BPSysAve, na.rm = TRUE),
    avg_dbp = mean(BPDiaAve, na.rm = TRUE)
  ) %>%
  mutate(across(where(is.numeric), ~round(., 1)))
```

```
# A tibble: 5 × 5
  age_group      n hypertension_prev avg_sbp avg_dbp
  <fct>      <dbl>          <dbl>   <dbl>   <dbl>
1 0-17      1009             0.4    104.    54.2
2 18-34     1446             3.1    113     67.1
3 35-49     1229            11.1    118     73.3
4 50-64     1069            18.3    125     72.8
5 65+        889            34.9    134     64
```


Multiple Group Comparisons

Diabetes prevalence by age, gender, and education:

```
nhanes_clean %>%
  filter(
    !is.na(age_group),
    !is.na(Gender),
    !is.na(education_level),
    !is.na(Diabetes)
  ) %>%
  group_by(age_group, Gender, education_level) %>%
  summarise(
    n = n(),
    diabetes_prev = mean(Diabetes == "Yes") * 100,
    .groups = "drop"
  ) %>%
  filter(n ≥ 20) %>% # Only groups with 20+ observations
  arrange(age_group, Gender, education_level) %>%
  head(10)
```

A tibble: 10 × 5

	age_group	Gender	education_level	n	diabetes_prev
	<fct>	<fct>	<fct>	<int>	<dbl>
1	18-34	female	Less than HS	108	0
2	18-34	female	High School	148	1.35

3	18-34	female	Some College	256	2.73
4	18-34	female	College+	166	1.20
5	18-34	male	Less than HS	128	1.56
6	18-34	male	High School	158	2.53
7	18-34	male	Some College	218	2.75
8	18-34	male	College+	147	2.04
9	35-49	female	Less than HS	108	11.1
10	35-49	female	High School	109	9.17

Creating Summary Tables

```
health_summary <- nhanes_clean %>%  
  filter(!is.na(Gender), !is.na(age_group)) %>%  
  group_by(Gender, age_group) %>%  
  summarise(  
    N = n(),  
    `Mean Age` = round(mean(Age, na.rm = TRUE), 1),  
    `Mean BMI` = round(mean(BMI, na.rm = TRUE), 1),  
    `Diabetes (%)` = round(mean(Diabetes == "Yes", na.rm = TRUE) * 100, 1),  
    `Hypertension (%)` = round(mean(hypertension == "Yes", na.rm = TRUE) * 100, 1),  
    .groups = "drop"  
  )
```

Health Summary Table

```
head(health_summary, 10)
```

```
# A tibble: 10 × 7
  Gender age_group      N `Mean Age` `Mean BMI` `Diabetes (%)`
  <fct>  <fct>    <int>    <dbl>    <dbl>    <dbl>
1 female 0-17      939      8.3     20.2      0.5
2 female 18-34     763     26.1     28.3      1.4
3 female 35-49     655     41.9     29.1      5.8
4 female 50-64     553     56.8     30.1     17.4
5 female 65+      510     73.8     29      23.6
6 male   0-17    1005      8.3     19.8      0.3
7 male   18-34     748     25.9     27.5       2
8 male   35-49     635     41.9     29.3      6.6
9 male   50-64     557     56.5     29.1     20.5
10 male   65+      414     73.2     28.3     26.3
# i 1 more variable: `Hypertension (%)` <dbl>
```

Calculating Prevalence Ratios

Example: Diabetes prevalence by education (reference: College+)

```
# Calculate prevalence by education
diabetes_by_ed <- nhanes_clean %>%
  filter(!is.na(education_level), !is.na(Diabetes)) %>%
  group_by(education_level) %>%
  summarise(
    prevalence = mean(Diabetes == "Yes") * 100
  )

# Calculate prevalence ratios
diabetes_by_ed %>%
  mutate(
    reference_prev = prevalence[education_level == "College+"],
    prevalence_ratio = prevalence / reference_prev
  ) %>%
  mutate(across(where(is.numeric), ~round(., 2)))
```

```
# A tibble: 4 × 4
  education_level prevalence reference_prev prevalence_ratio
<fct>           <dbl>         <dbl>         <dbl>
1 Less than HS    17.3           7.95           2.18
2 High School     11.2           7.95           1.41
```

Age Standardization Example

Simple direct standardization:

```
# Reference population (all ages)
reference_age_dist <- nhanes_clean %>%
  count(age_group) %>%
  mutate(weight = n / sum(n))

# Age-specific rates by gender
age_specific_rates <- nhanes_clean %>%
  filter(!is.na(Diabetes), !is.na(age_group)) %>%
  group_by(Gender, age_group) %>%
  summarise(
    rate = mean(Diabetes == "Yes") * 100,
    .groups = "drop"
  )

# Standardized rates
age_specific_rates %>%
  left_join(reference_age_dist, by = "age_group") %>%
  group_by(Gender) %>%
  summarise(
    crude_rate = mean(rate),
    standardized_rate = sum(rate * weight)
```

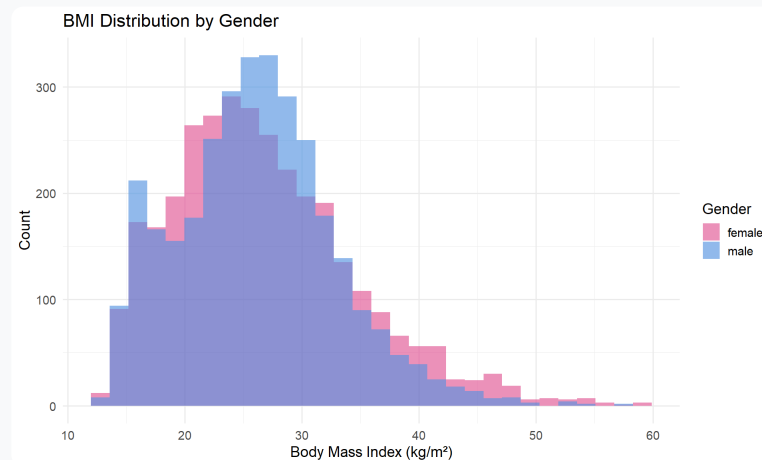
```
# A tibble: 2 × 3
  Gender crude_rate standardized_rate
  <fct>      <dbl>          <dbl>
1 female      9.73            7.61
2 male      11.2            8.74
```


Population Pyramid

```
nhanes_clean %>%
  filter(!is.na(age_group), !is.na(Gender)) %>%
  count(age_group, Gender) %>%
  mutate(
    n = ifelse(Gender == "male", -n, n)
  ) %>%
  ggplot(aes(x = age_group, y = n, fill = Gender)) +
  geom_col() +
  coord_flip() +
  scale_y_continuous(
    labels = abs,
    name = "Population Count"
  ) +
  scale_fill_manual(values = c("male" = "#4A90E2", "female" = "#E24A90")) +
  labs(
    title = "Population Pyramid: NHANES Sample",
    x = "Age Group",
    fill = "Gender"
  ) +
  theme_minimal(base_size = 14) +
  theme(legend_position = "bottom")
```

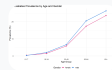

BMI Distribution

```
nhanes_clean %>%  
  filter(!is.na(BMI), BMI < 60) %>%  
  ggplot(aes(x = BMI, fill = Gender)) +  
  geom_histogram(bins = 30, alpha = 0.6, position = "identity") +  
  scale_fill_manual(values = c("male" = "#4A90E2", "female" = "#E24A90")) +  
  labs(  
    title = "BMI Distribution by Gender",  
    x = "Body Mass Index (kg/m²)",  
    y = "Count"  
  ) +  
  theme_minimal(base_size = 14)
```



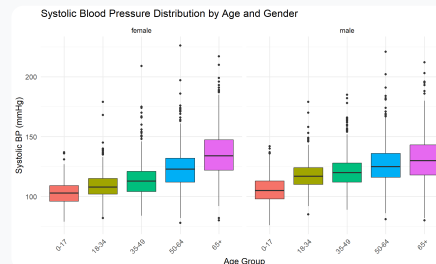
Diabetes Prevalence by Age

```
nhanes_clean %>%  
  filter(!is.na(age_group), !is.na(Diabetes)) %>%  
  group_by(age_group, Gender) %>%  
  summarise(  
    prevalence = mean(Diabetes == "Yes") * 100,  
    .groups = "drop"  
  ) %>%  
  ggplot(aes(x = age_group, y = prevalence,  
             color = Gender, group = Gender)) +  
  geom_line(size = 1.2) +  
  geom_point(size = 3) +  
  scale_color_manual(values = c("male" = "#4A90E2", "female" = "#E24A90")) +  
  labs(  
    title = "Diabetes Prevalence by Age and Gender",  
    x = "Age Group",  
    y = "Prevalence (%)"  
  ) +  
  theme_minimal(base_size = 14) +  
  theme(legend.position = "bottom")
```



Blood Pressure by Age

```
nhanes_clean %>%  
  filter(!is.na(age_group), !is.na(BPSysAve)) %>%  
  ggplot(aes(x = age_group, y = BPSysAve, fill = age_group)) +  
  geom_boxplot() +  
  facet_wrap(~Gender) +  
  labs(  
    title = "Systolic Blood Pressure Distribution by Age and Gender",  
    x = "Age Group",  
    y = "Systolic BP (mmHg)"  
  ) +  
  theme_minimal(base_size = 14) +  
  theme(  
    legend.position = "none",  
    axis.text.x = element_text(angle = 45, hjust = 1)  
  )
```



Education and Health

```
nhanes_clean %>%
  filter(!is.na(education_level), !is.na(bmi_category)) %>%
  count(education_level, bmi_category) %>%
  group_by(education_level) %>%
  mutate(percentage = n / sum(n) * 100) %>%
  ggplot(aes(x = education_level, y = percentage, fill = bmi_category)) +
  geom_col(position = "dodge") +
  scale_fill_brewer(palette = "Set2") +
  labs(
    title = "BMI Categories by Education Level",
    x = "Education Level",
    y = "Percentage (%)",
    fill = "BMI Category"
  ) +
  theme_minimal(base_size = 14) +
  theme(
    axis.text.x = element_text(angle = 45, hjust = 1),
    legend.position = "bottom"
  )
```



Exercise 1: Basic Analysis

Task: Calculate obesity prevalence by race/ethnicity

```
# Your code here
nhanes_clean %>%
  filter(!is.na(___), !is.na(___)) %>%
  group_by(___) %>%
  summarise(
    n = ___,
    obesity_prev = mean(___ == "___") * 100
  )
```

Exercise 1: Solution

```
nhanes_clean %>%  
  filter(!is.na(Race1), !is.na(bmi_category)) %>%  
  group_by(Race1) %>%  
  summarise(  
    n = n(),  
    obesity_prev = mean(bmi_category == "Obese") * 100  
  ) %>%  
  arrange(desc(obesity_prev)) %>%  
  mutate(obesity_prev = round(obesity_prev, 1))
```

```
# A tibble: 5 × 3  
  Race1      n obesity_prev  
  <fct>   <int>         <dbl>  
1 Black     987          37.7  
2 Mexican   808          28.3  
3 White  3614           27  
4 Hispanic  475          26.9  
5 Other    591          15.7
```

Exercise 2: Multiple Variables

Task: Analyze physical activity by age group and poverty status

```
# Your code here
nhanes_clean %>%
  filter(!is.na(___), !is.na(___), !is.na(___)) %>%
  group_by(___, ___) %>%
  summarise(
    active_pct = ___,
    .groups = "drop"
  )
```


Exercise 2: Solution

```
nhanes_clean %>%
  filter(
    !is.na(age_group),
    !is.na(poverty_status),
    !is.na(PhysActive)
  ) %>%
  group_by(age_group, poverty_status) %>%
  summarise(
    n = n(),
    active_pct = mean(PhysActive == "Yes") * 100,
    .groups = "drop"
  ) %>%
  filter(n ≥ 20) %>%
  mutate(active_pct = round(active_pct, 1)) %>%
  arrange(age_group, poverty_status)
```

A tibble: 15 × 4

	age_group <fct>	poverty_status <chr>	n <int>	active_pct <dbl>
1	0-17	Above poverty	304	85.9
2	0-17	Below poverty	129	75.2
3	0-17	Near poverty	135	80
4	18-34	Above poverty	637	70.5
5	18-34	Below poverty	400	53.8
6	18-34	Near poverty	343	58
7	35-49	Above poverty	716	63.3

8	35-49	Below poverty	213	39.9
9	35-49	Near poverty	273	41.4
10	50-64	Above poverty	692	54.3
11	50-64	Below poverty	147	26.5
12	50-64	Near poverty	179	36.3
13	65+	Above poverty	512	44.9
14	65+	Below poverty	104	21.2
15	65+	Near poverty	227	29.1

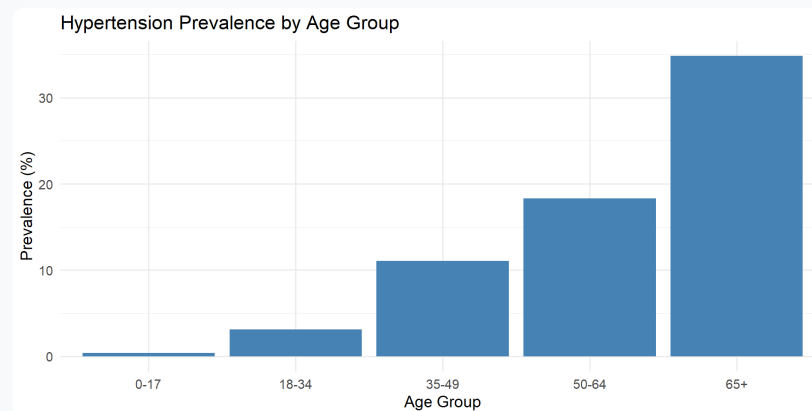
Exercise 3: Create Visualization

Task: Create a bar chart of hypertension prevalence by age group

```
# Your code here
nhanes_clean %>%
  filter(!is.na(___), !is.na(___)) %>%
  group_by(___) %>%
  summarise(prev = ___) %>%
  ggplot(aes(x = ___, y = ___)) +
  geom_col(fill = "steelblue") +
  labs(title = "___")
```

Exercise 3: Solution

```
nhanes_clean %>%  
  filter(!is.na(age_group), !is.na(hypertension)) %>%  
  group_by(age_group) %>%  
  summarise(prev = mean(hypertension == "Yes") * 100) %>%  
  ggplot(aes(x = age_group, y = prev)) +  
  geom_col(fill = "steelblue") +  
  labs(  
    title = "Hypertension Prevalence by Age Group",  
    x = "Age Group",  
    y = "Prevalence (%)"  
  ) +  
  theme_minimal(base_size = 14)
```



Exercise 4: Complex Analysis

Task: Compare mean BMI across education levels, separately for each gender and age group (adults only)

Hints:

1. Filter for adults (Age ≥ 18)
2. Remove missing values
3. Group by three variables
4. Calculate mean BMI
5. Filter for groups with $n \geq 10$

Exercise 4: Solution

```
nhanes_clean %>%
  filter(
    Age ≥ 18,
    !is.na(education_level),
    !is.na(Gender),
    !is.na(age_group),
    !is.na(BMI)
  ) %>%
  group_by(Gender, age_group, education_level) %>%
  summarise(
    n = n(),
    mean_bmi = mean(BMI),
    .groups = "drop"
  ) %>%
  filter(n ≥ 10) %>%
  mutate(mean_bmi = round(mean_bmi, 1)) %>%
  arrange(Gender, age_group, education_level) %>%
  head(12)
```

A tibble: 12 × 5

	Gender	age_group	education_level	n	mean_bmi
	<fct>	<fct>	<fct>	<int>	<dbl>
1	female	18-34	Less than HS	108	30
2	female	18-34	High School	147	30.5
3	female	18-34	Some College	251	28.5

4	female	18-34	College+	166	26.1
5	female	35-49	Less than HS	107	30.4
6	female	35-49	High School	108	29.9
7	female	35-49	Some College	193	30
8	female	35-49	College+	238	27.4
9	female	50-64	Less than HS	111	30.9
10	female	50-64	High School	113	31.3
11	female	50-64	Some College	165	30.9
12	female	50-64	College+	160	27.9

Saving Cleaned Data

```
# Save as CSV
export(nhanes_clean, "nhanes_cleaned.csv")

# Save as RDS (preserves R object types)
export(nhanes_clean, "nhanes_cleaned.rds")

# Save as Stata file
export(nhanes_clean, "nhanes_cleaned.dta")

# Save as Excel
export(nhanes_clean, "nhanes_cleaned.xlsx")
```

Exporting Summary Tables

```
# Create summary
summary_table <- nhanes_clean %>%
  group_by(Gender, age_group) %>%
  summarise(
    N = n(),
    `Mean BMI` = mean(BMI, na.rm = TRUE),
    `Diabetes (%)` = mean(Diabetes == "Yes", na.rm = TRUE) * 100,
    .groups = "drop"
  )

# Export
export(summary_table, "health_summary.csv")
export(summary_table, "health_summary.xlsx")
```

Saving Plots

```
# Create plot
p <- nhanes_clean %>%
  filter(!is.na(age_group), !is.na(Diabetes)) %>%
  group_by(age_group) %>%
  summarise(prev = mean(Diabetes == "Yes") * 100) %>%
  ggplot(aes(x = age_group, y = prev)) +
  geom_col(fill = "steelblue") +
  labs(title = "Diabetes Prevalence by Age")

# Save
ggsave("diabetes_by_age.png", p, width = 10, height = 6, dpi = 300)
ggsave("diabetes_by_age.pdf", p, width = 10, height = 6)
```


Data Quality Checks

1. Check for missing data

- Use `summary()` or `datasummary_skim()`
- Decide on handling strategy (remove, impute, analyze separately)

2. Verify coding

- Check factor levels: `levels(data$variable)`
- Confirm numeric ranges: `range(data$variable, na.rm = TRUE)`

3. Look for outliers

- Use boxplots and summary statistics
- Investigate extreme values

4. Check sample sizes

- Ensure adequate n in each group
- Consider combining categories if needed

Analytical Best Practices

1. **Age-standardize when comparing populations**
 - Different age structures can confound comparisons
2. **Report confidence intervals**
 - Not just point estimates
3. **Consider survey weights**
 - If using complex survey data
4. **Document your decisions**
 - Keep code comments
 - Note any exclusions or transformations
5. **Create reproducible workflows**
 - Use R scripts or Quarto documents
 - Include session info: `sessionInfo()`

Visualization Best Practices

1. Choose appropriate chart types

- Population pyramids for age-sex structure
- Line charts for trends
- Bar charts for comparisons

2. Use clear labels

- Title, axis labels, legend
- Include sample size where relevant

3. Color considerations

- Use colorblind-friendly palettes
- Avoid red-green combinations

4. Keep it simple

- One clear message per chart
- Avoid chartjunk

Common Pitfalls to Avoid

Watch Out For:

- ✗ **Simpson's Paradox** - Always consider confounders
- ✗ **Small sample sizes** - $n < 30$ can be unreliable
- ✗ **Multiple testing** - Adjust p-values if doing many tests
- ✗ **Correlation $\neq$ Causation** - Be careful with language
- ✗ **Selection bias** - Who's included/excluded?
- ✗ **Ignoring missing data** - Can introduce bias
- ✗ **Over-interpretation** - Don't overstate findings

Demographic Research Questions

Population Health:

- What is the obesity prevalence in different age groups?
- How does diabetes vary by socioeconomic status?
- Are there gender differences in cardiovascular risk?

Health Inequalities:

- Do education gradients in health exist?
- How do health outcomes vary by race/ethnicity?
- What is the urban-rural health gap?

Policy Relevant:

- Which populations need targeted interventions?
- How effective are prevention programs?
- What are priority health issues for different age groups?

Example: Health Inequality Report

Research Question: *How does diabetes prevalence vary by education and poverty status?*

```
diabetes_inequality <- nhanes_clean %>%  
  filter(  
    Age ≥ 25, # Adults who completed education  
    !is.na(education_level),  
    !is.na(poverty_status),  
    !is.na(Diabetes)  
  ) %>%  
  group_by(education_level, poverty_status) %>%  
  summarise(  
    n = n(),  
    diabetes_prev = mean(Diabetes == "Yes") * 100,  
    .groups = "drop"  
  ) %>%  
  filter(n ≥ 20)
```

Health Inequality Results

```
diabetes_inequality %>%  
  arrange(education_level, poverty_status) %>%  
  mutate(diabetes_prev = round(diabetes_prev, 1))
```

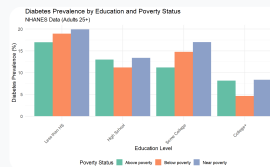
A tibble: 12 × 4

	education_level <fct>	poverty_status <chr>	n <int>	diabetes_prev <dbl>
1	Less than HS	Above poverty	236	16.9
2	Less than HS	Below poverty	280	18.9
3	Less than HS	Near poverty	286	19.9
4	High School	Above poverty	416	13
5	High School	Below poverty	161	11.2
6	High School	Near poverty	224	13.4
7	Some College	Above poverty	733	11.2
8	Some College	Below poverty	149	14.8
9	Some College	Near poverty	253	17
10	College+	Above poverty	968	8.2
11	College+	Below poverty	43	4.7
12	College+	Near poverty	120	8.3

Key Finding: Clear socioeconomic gradient in diabetes prevalence

Visualizing Health Inequality

```
diabetes_inequality %>%  
  ggplot(aes(x = education_level, y = diabetes_prev,  
             fill = poverty_status)) +  
  geom_col(position = "dodge") +  
  scale_fill_brewer(palette = "Set2") +  
  labs(  
    title = "Diabetes Prevalence by Education and Poverty Status",  
    subtitle = "NHANES Data (Adults 25+)",  
    x = "Education Level",  
    y = "Diabetes Prevalence (%)",  
    fill = "Poverty Status"  
  ) +  
  theme_minimal(base_size = 14) +  
  theme(  
    axis.text.x = element_text(angle = 45, hjust = 1),  
    legend.position = "bottom"  
  )
```



Project Goal

Create a comprehensive cardiovascular risk profile by demographic groups

Variables to analyze:

- Hypertension (high blood pressure)
- Obesity ($\text{BMI} \geq 30$)
- Diabetes
- Physical inactivity
- Smoking

Stratify by: Age, Sex, Education

Step 1: Create Risk Indicators

```
cv_risk_data <- nhanes_clean %>%  
  filter(Age ≥ 18) %>%  
  mutate(  
    has_hypertension = hypertension == "Yes",  
    has_obesity = bmi_category == "Obese",  
    has_diabetes = Diabetes == "Yes",  
    inactive = PhysActive == "No",  
    current_smoker = SmokeNow == "Yes",  
  
    risk_count = rowSums(  
      across(c(has_hypertension, has_obesity, has_diabetes,  
                inactive, current_smoker), ~ as.numeric(.)),  
      na.rm = TRUE  
    )  
  )
```

Step 2: Calculate Risk Prevalence

```
cv_risk_summary <- cv_risk_data %>%  
  filter(!is.na(age_group), !is.na(Gender)) %>%  
  group_by(age_group, Gender) %>%  
  summarise(  
    n = n(),  
    hypertension_pct = mean(has_hypertension, na.rm = TRUE) * 100,  
    obesity_pct = mean(has_obesity, na.rm = TRUE) * 100,  
    diabetes_pct = mean(has_diabetes, na.rm = TRUE) * 100,  
    inactive_pct = mean(inactive, na.rm = TRUE) * 100,  
    smoking_pct = mean(current_smoker, na.rm = TRUE) * 100,  
    mean_risk_count = mean(risk_count, na.rm = TRUE),  
    .groups = "drop"  
  ) %>%  
  mutate(across(where(is.numeric), ~round(., 1)))
```

Step 3: Results

```
cv_risk_summary %>%  
  arrange(age_group, Gender) %>%  
  head(10)
```

```
# A tibble: 8 × 9  
  age_group Gender      n hypertension_pct obesity_pct diabetes_pct inactive_pct  
  <fct>      <fct> <dbl>          <dbl>         <dbl>         <dbl>         <dbl>  
1 18-34     female  763            1.5          32.7           1.4          44.2  
2 18-34     male    748            4.7          28.7           2            32.2  
3 35-49     female  655            7.8          38.1           5.8          48.7  
4 35-49     male    635           14.3          36.9           6.6          43.9  
5 50-64     female  553            15           43.2          17.4          53  
6 50-64     male    557           21.6          35.8          20.5          56  
7 65+       female  510           37.7          37.6          23.6          67.8  
8 65+       male    414           31.4          32.9          26.3          56.5  
# i 2 more variables: smoking_pct <dbl>, mean_risk_count <dbl>
```

Step 4: Visualization

```
cv_risk_summary %>%
  select(age_group, Gender, hypertension_pct, obesity_pct,
         diabetes_pct, inactive_pct) %>%
  pivot_longer(
    cols = ends_with("_pct"),
    names_to = "risk_factor",
    values_to = "prevalence"
  ) %>%
  mutate(
    risk_factor = str_remove(risk_factor, "_pct"),
    risk_factor = str_to_title(risk_factor)
  ) %>%
  ggplot(aes(x = age_group, y = prevalence,
            color = risk_factor, group = risk_factor)) +
  geom_line(linewidth = 1) +
  geom_point(size = 2) +
  facet_wrap(~Gender) +
  labs(
    title = "Cardiovascular Risk Factors by Age and Gender",
    x = "Age Group",
    y = "Prevalence (%)"
  )
```


Key R Packages for Demographic Analysis

Data Wrangling:

- `tidyverse` - Complete data science toolkit
- `rio` - Universal data import/export
- `janitor` - Data cleaning helpers

Survey Analysis:

- `survey` - Complex survey design analysis
- `srvyr` - Survey data + dplyr syntax
- `weights` - Weighted statistics

Demographic Analysis:

- `DemoTools` - Demographic methods
- `demography` - Mortality/fertility
- `popbio` - Population biology

Learning Resources

Free Books:

- [R for Data Science](#)
- [Advanced R](#)
- [ggplot2 book](#)

Demographic Analysis:

- [IPUMS training](#)
- [DHS Program tutorials](#)
- [WHO indicators guide](#)

Practice:

- [Tidy Tuesday](#)
- [Our World in Data](#)
- [NHANES tutorials](#)

What We Covered Today

- ✓ Importing and exploring demographic/health data
- ✓ Data cleaning and transformation
- ✓ Creating demographic indicators
- ✓ Age-sex analysis patterns
- ✓ Socioeconomic gradients
- ✓ Data visualization
- ✓ Reproducible workflows
- ✓ Real-world applications

Next Steps in Your Learning

1. Practice with real data

- Download DHS, MICS, or census data
- Replicate published analyses

2. Learn visualization

- Master ggplot2
- Create publication-quality figures

3. Statistical analysis

- Regression models
- Survival analysis
- Multilevel models

4. Advanced topics

- Survey weights
- Missing data methods

• Age standardization

Your Workshop Toolkit

Files you'll receive:

-  This slide deck (.qmd)
-  Practice workbook (.Rmd)
-  Sample NHANES dataset (.csv)
-  Complete code scripts (.R)
-  Exercise solutions
-  Resource list

All materials are reproducible and yours to keep!

Thank You!

Questions?

Contact & Resources:

- Workshop materials: [GitHub repository]
- Email: [your email]
- Further learning: [Resource links]

Remember

The best way to learn R is by doing. Start with small projects, build confidence, and gradually tackle more complex analyses!

"In God we trust. All others must bring data." - W. Edwards Deming